Physics-informed residual learning of descriptor power-network models: prediction, contingency margins and a control test

1 Model

Swing dynamics and power balance kept

0 = g(x, z, u): algebraic angles

Only the missing differential term is learned (weak-form sparse regression)

Newton projection returns the bus angles to the constraint manifold at every stage

2 Results (lossless case9, case39)

Mean per-run state-RMSE ratio (10 horizons): 0.17 to 0.24 of physics-only

Algebraic residual at solver tolerance

Row-stochastic angle sensitivity: algebraic error <= rotor-angle span

Analytic gradient: case39 control step 588 ms → 24 ms

Case39: 3.6-4.2% gain vs static law; no consistent gain over recalibrated physics

3 Outage certificate and margin

Outage certificate: Newton-Kantorovich, held on all 41 non-islanding outages

Predictor rerun under outage topology: pooled coverage 0.945, 4 calibration draws

Misses lines 8, 10, 11, 37; unresolved

Coverage conditional on exchangeability

No independent AC, real-grid or stability validation